SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

# ACB 2000A (ACB-AS-2000) — Datasheet

## 1. Overview

Air circuit breaker 2000A, draw-out type, electronic trip unit (LSIG). Product family: Low-voltage circuit breakers. Lifecycle status: **ACTIVE**.

## 2. Technical specifications

### 2.1. Ratings

| Item | Value |
|---|---|
| Rated current (In) | 2000 A |

### 2.2. Performance and characteristics

| Item | Value |
|---|---|
| Ultimate breaking capacity Icu at AC 415 V | 85 kA |
| Short-time withstand current Icw (1 s) | 65 kA |

### 2.3. Endurance and communication

| Item | Value |
|---|---|
| Closing time | ≤ 70 ms |
| Mechanical endurance (with maintenance) | 20,000 operations |

### 2.4. Protection and options

| Item | Value |
|---|---|
| Trip unit protection functions | L (long-time), S (short-time), I (instantaneous), G (ground fault) |

### 2.5. Monitoring and auxiliaries

| Item | Value |
|---|---|
| Motor operator control voltage | AC/DC 110-240 V |
| Trip unit communication | Modbus RTU (RS-485) |

## 3. Installation data

### 3.1. Terminals

- Dimensions (W x H x D): 420 x 430 x 380 (draw-out) mm
- Weight: 78 kg

- Degree of protection: IP31 (with door frame)
- Operating temperature: -5 to +55 °C

## 4. Standards and certifications

Applicable standards:

- IEC 60947-2

Certifications held:

- CE
- KC
- CB

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

## 5. Ordering information

| Item | Value |
|---|---|
| Ordering code | ACB-AS-2000 |
| Standard lead time | 8 weeks ex-works |
| Minimum order quantity | 1 |
| Warranty | 24 months (see POL-WARRANTY) |
| Firmware | latest 1.4 (see RN-ACB-AS-2000) |

## 6. Related documents

- IG-ACB-AS-2000 — Installation and Commissioning Guide
- TG-ACB-AS-2000 — Troubleshooting and Error Code Guide
- RN-ACB-AS-2000 — Firmware Release Notes
- FAQ-LV_BREAKER — Low-voltage circuit breakers FAQ

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# ACB 2000A (ACB-AS-2000) — Installation and Commissioning Guide

## 1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

## 2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

## 3. Mounting and clearances

- Required clearances: 50 mm above and below the breaker for arc gases; 10 mm between adjacent breakers
- Operating temperature: -5 to +55 °C
- Degree of protection: IP31 (with door frame)
- Above 1,000 m altitude, apply derating per the datasheet.

## 4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

## 5. Commissioning procedure

1. Check the rating plate against the single-line diagram
2. Verify tightening torque of all terminals
3. Measure insulation resistance (≥ 5 MΩ at 500 V DC)
4. Operate the breaker manually ON/OFF three times
5. Check trip unit settings against the coordination study

## 6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

## 7. Common commissioning issues

| Symptom | Likely cause | See |
|---|---|---|
| Terminal overheating / discoloration | Loose connection or undersized cable | TG-ACB-AS-2000 §3.1 |
| ACB long-time trip | Sustained overload above long-time setting | TG-ACB-AS-2000 §3.2 |
| ACB ground fault trip | Insulation fault in downstream feeder | TG-ACB-AS-2000 §3.3 |
| ACB trip unit self-diagnosis alarm | Firmware defect after control power dip | TG-ACB-AS-2000 §3.4 |

On-site commissioning support can be requested through the Global CS Center. For low-voltage circuit breakers the request is handled by the technical support team.

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# ACB 2000A (ACB-AS-2000) — Firmware Release Notes

## 1. Summary

Latest firmware: **1.4**. Updating to the latest version is recommended for all units.

## 2. Version history

### 2.1. Version 1.4 (2026-03-09)

- Fixed: false T.SF self-diagnosis alarm after control power dip below 85 V
- Improved: event log capacity from 64 to 256 entries

### 2.2. Version 1.3 (2025-06-16)

- Fixed: ground fault (G) pickup delay ignored when set to 0.1 s
- Added: Modbus register for contact wear percentage

### 2.3. Version 1.2 (2024-11-04)

- Initial release of trip unit firmware for AS series

## 3. Update procedure

1. Back up parameters / project before updating.
2. Connect the engineering tool (v4.6 or later).
3. Do not switch off power during the update.
4. After the update, verify the version and restore parameters if required.
5. If the update fails, do not power-cycle the unit repeatedly; contact support for the recovery procedure.

## 4. Compatibility

Parameter files from older versions can be loaded; files from newer versions cannot be loaded into older firmware.